Experimentation & Evaluation

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Lecture 06

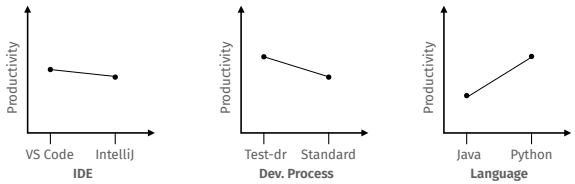
Experimental Design

More than one independent variable

I want to test the impact on developers' productivity of (i) using different IDEs (first factor), (ii) adopting different development processes (e.g., test-driven vs standard), and (iii) programming in different languages.

Test one-factor-at-time (OFAT)

Test different IDEs first (by keeping constant the rest) development processes then and finally programming languages.



More than one independent variable

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Full factorial design

Test combinations of factors
k factors each having n levels = n^k tests!!!

VSCode	TDD	Java
VSCode	TDD	Python
VSCode	Standard	Java
VSCode	Standard	Python
IntelliJ	TDD	Java
IntelliJ	TDD	Python
IntelliJ	Standard	Java
IntelliJ	Standard	Python

Fractional factorial design

To be used when there are too many factors
E.g., Half-fractional would test 4 out of 8.

The more we reduce, the less we can analyze interactions.

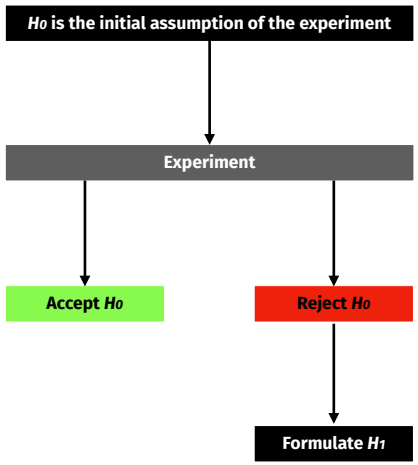
VSCode	TDD	Java
VSCode	Standard	Python
IntelliJ	TDD	Python
IntelliJ	Standard	Java

Null Hypothesis

Null Hypothesis

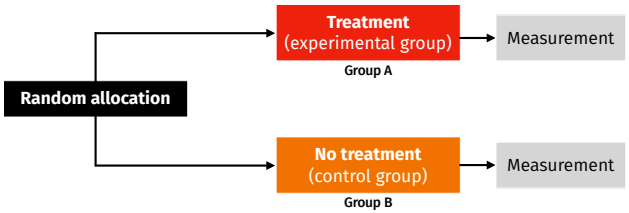
I want to test the impact on developers' productivity of adopting different development processes (e.g., test-driven vs standard).

Null Hypothesis (H_0): *There is no statistically significant difference between the productivity of developers when using TDD and the standard development process.*

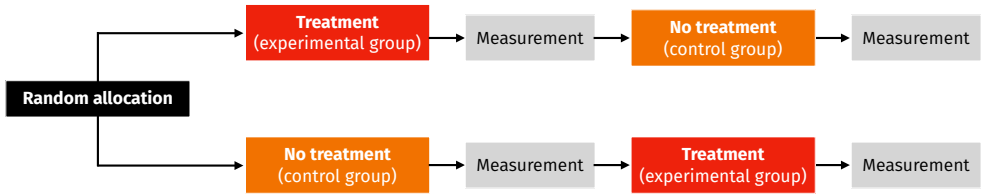




Between groups design



Within subjects design



I developed a new source code recommender system (e.g., GitHub Copilot) and I want to test its impact on developers' productivity.

Do they produce "better" code (e.g., fewer bugs) maybe even faster?

Within subjects design

How to avoid order effect?

	All participants
Lab 1	With Code Recommender (WCR), Task T1
Lab 2	No Code Recommender (NCR), Task T2

Within subjects design

How to avoid order effect?

	Group A	Group B
Lab 1	WCR T1	NCR T2
Lab 2	NCR T2	WCR T1

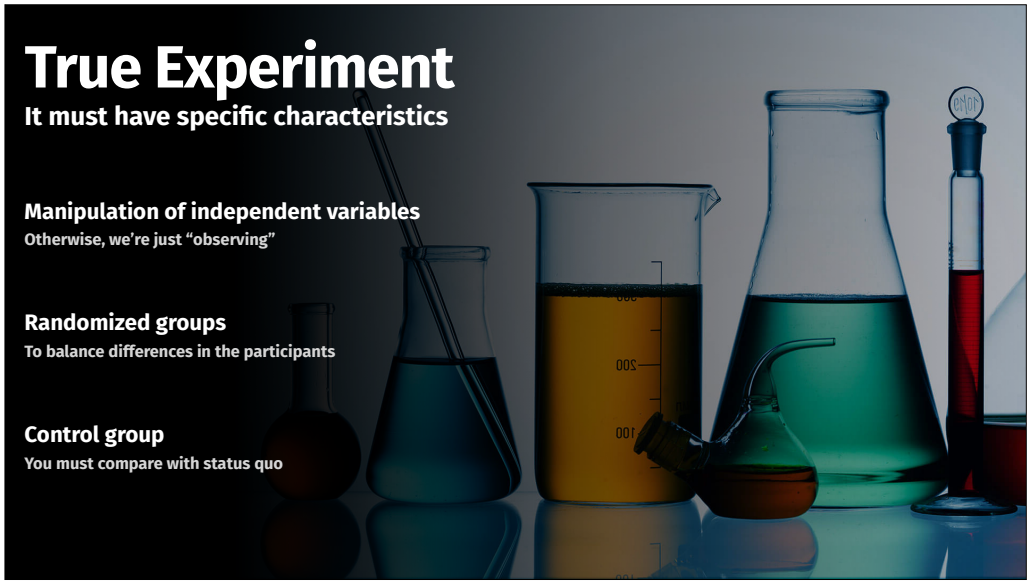
Latin Square arrangement

Within subjects design

What about experimental material?

	Group A	Group B	Group C	Group D
Lab 1	WCR T1	NCR T2	WCR T2	NCR T1
Lab 2	NCR T2	WCR T1	NCR T1	WCR T2

Latin Square arrangement



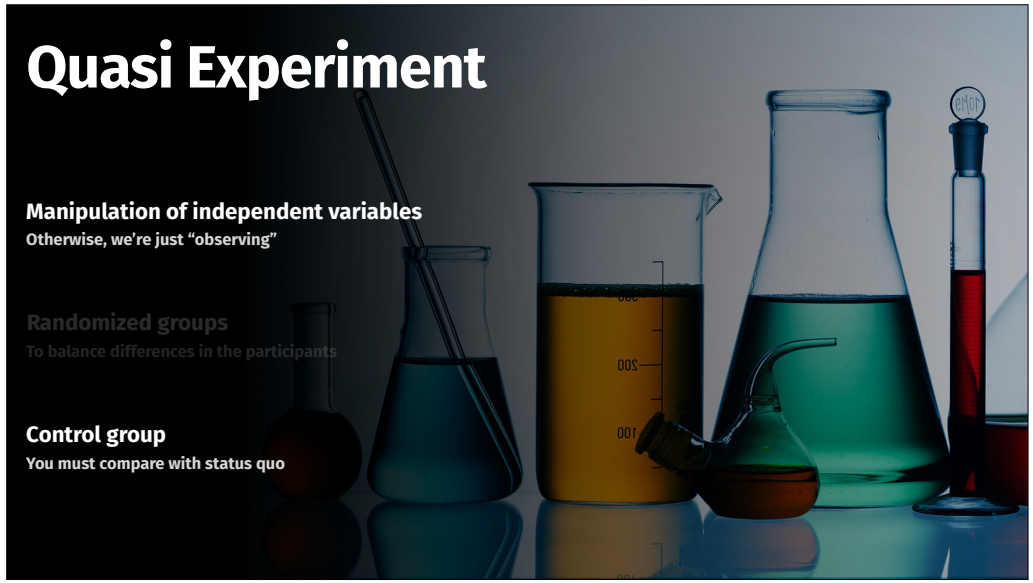
True Experiment

It must have specific characteristics

Manipulation of independent variables
Otherwise, we're just "observing"

Randomized groups
To balance differences in the participants

Control group
You must compare with status quo



Quasi Experiment

Manipulation of independent variables
Otherwise, we're just "observing"

Randomized groups
To balance differences in the participants

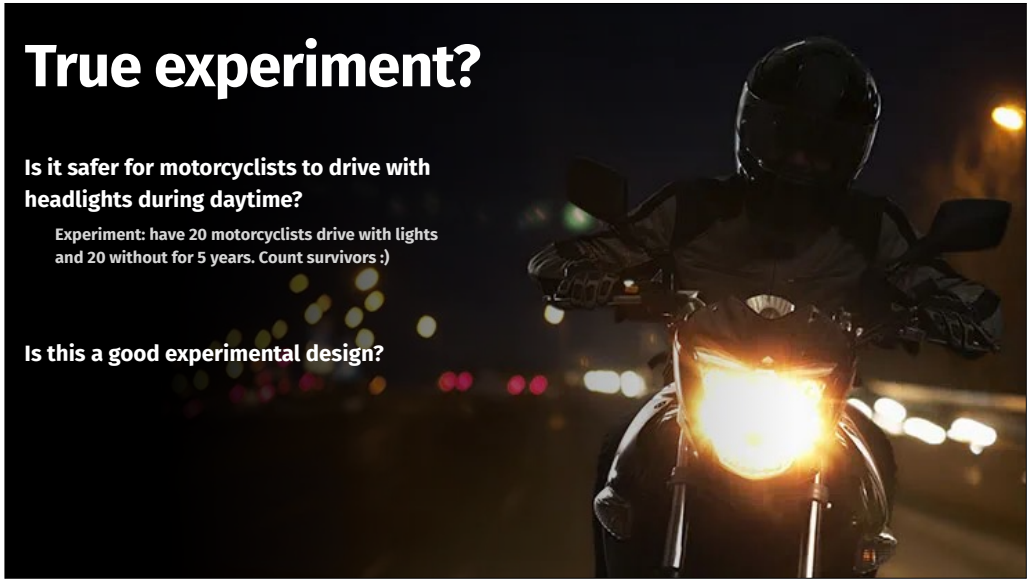
Control group
You must compare with status quo

True experiment?

Is it safer for motorcyclists to drive with headlights during daytime?

Experiment: have 20 motorcyclists drive with lights and 20 without for 5 years. Count survivors :)

Is this a good experimental design?



True experiment?

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Experiment: have 20 motorcyclists drive with lights and 20 without for 5 years. Count survivors :)

Is this a good experimental design?

Real problem here is ethics! I cannot run this study.

But I can run a quasi-experiment. I just observe people that by their choice use/not use lights during daytime.



Beyond Experiments

Quasi experiment
No randomized groups

Survey
Self-reports instead of measurements.
Problem: selection bias.

Case study
Observation of participants over long period of time. Usually hard to generalize.

Naturalistic observations
No awareness of participants that they're studied.

